Class 9 Mathematics — Exercise 4.6

Basic Concept

Operations on Algebraic Fractions

- **Multiplication:** Factorize the numerators and denominators, cancel the common factors, then multiply.
- **Division:** Change $\div$ into $\times$ and take the **reciprocal** (turn upside down) of the fraction after the $\div$ sign.

$$\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \times \frac{D}{C}$$

- **Addition / Subtraction:** Factorize the denominators, take the **LCM** of the denominators, then add or subtract the numerators.
- **Reciprocal:** The product of a fraction and its reciprocal is always 1: $\frac{A}{B} \times \frac{B}{A} = 1$

Question 1

Answer these without calculations.

(i) Product of what algebraic fraction and $x^3 + 7x - 8$ is 1?

$$\boxed{\frac{1}{x^3 + 7x - 8}} \qquad \left(\because (x^3 + 7x - 8) \times \frac{1}{x^3 + 7x - 8} = 1\right)$$

(ii) Which algebraic fraction divided by $\frac{x^2}{x^2 + y^2}$ gives 1?

$$\boxed{\frac{x^2}{x^2 + y^2}} \qquad (\because \text{a fraction divided by itself is 1})$$

(iii) Sum of what algebraic fraction and $\frac{m}{m^2 + n^2}$ is $\frac{m + n}{m^2 + n^2}$?

$$\boxed{\frac{n}{m^2 + n^2}} \qquad \left(\because \frac{m}{m^2 + n^2} + \frac{n}{m^2 + n^2} = \frac{m + n}{m^2 + n^2}\right)$$

(iv) What is the product of an algebraic fraction and its reciprocal?

$$\boxed{1} \qquad \left(\text{e.g. } 5 \times \frac{1}{5} = 1\right)$$

Question 2

$$\frac{14x^2 - 7x}{12x^3 + 24x^2} \times \frac{x^2 + 2x}{2x - 1}$$

$$= \frac{7x(2x-1)}{12x^2(x+2)} \times \frac{x(x+2)}{2x-1}$$

$$= \frac{7x^2}{12x^2}$$

$$= \boxed{\frac{7}{12}} \text{ Ans}$$

Question 3

$$\frac{a^2b^2+3ab}{4a^2-1} \times \frac{2a+1}{ab+3}$$

$$= \frac{ab(ab+3)}{(2a)^2-(1)^2} \times \frac{2a+1}{ab+3}$$

$$= \frac{ab(ab+3)}{(2a+1)(2a-1)} \times \frac{2a+1}{ab+3}$$

$$= \boxed{\frac{ab}{2a-1}} \text{ Ans}$$

Question 4

$$\frac{a-b}{a^2+ab} \times \frac{a^4-b^4}{a^2-2ab+b^2} \times \frac{a}{a^2+b^2}$$

$(\because (a-b)^2 = a^2-2ab+b^2)$

$$= \frac{a-b}{a(a+b)} \times \frac{(a^2)^2-(b^2)^2}{(a-b)^2} \times \frac{a}{a^2+b^2}$$

$$= \frac{a-b}{a(a+b)} \times \frac{(a^2+b^2)(a+b)(a-b)}{(a-b)(a-b)} \times \frac{a}{a^2+b^2}$$

$$= \boxed{1} \text{ Ans}$$

Question 5

$$\frac{6x^2y^2}{x^2-y^2} \div \frac{3xy}{x+y}$$

$$= \frac{6x^2y^2}{(x+y)(x-y)} \times \frac{x+y}{3xy}$$

$$= \boxed{\frac{2xy}{x-y}} \text{ Ans}$$

Question 6

$$\frac{8a^3-1}{4a^3+2a^2} \div \frac{6a^2-13a+5}{15a-25} \times \frac{2a^4+a^3}{15a^2}$$

For $6a^2-13a+5$: $30 \to -10$ and -3 $\quad(-10-3=-13)$

$$6a^2-13a+5 = 6a^2-10a-3a+5 = 2a(3a-5)-1(3a-5) = (3a-5)(2a-1)$$

$$= \frac{(2a)^3-(1)^3}{2a^2(2a+1)} \times \frac{15a-25}{6a^2-13a+5} \times \frac{a^3(2a+1)}{15a^2}$$

$$= \frac{(2a-1)[(2a)^2+(2a)(1)+(1)^2]}{2a^2(2a+1)} \times \frac{5(3a-5)}{(3a-5)(2a-1)} \times \frac{a^3(2a+1)}{15a^2}$$

$$= \frac{(4a^2+2a+1)\times 5 \times a^3}{2a^2 \times 15a^2}$$

$$= \boxed{\frac{4a^2+2a+1}{6a}} \text{ Ans}$$

Question 7

$$\frac{x^2-8x-9}{x^2-17x+72} \times \frac{x^2-25}{x^2-1} \div \frac{x^2+4x-5}{x^2-9x+8}$$

Rough work: $-9 \to -9, 1$; $\quad 72 \to -9, -8$; $\quad -5 \to 5, -1$; $\quad 8 \to -8, -1$

$$= \frac{x^2-9x+x-9}{x^2-9x-8x+72} \times \frac{(x)^2-(5)^2}{(x)^2-(1)^2} \div \frac{x^2+5x-x-5}{x^2-8x-x+8}$$

$$= \frac{x(x-9)+1(x-9)}{x(x-9)-8(x-9)} \times \frac{(x+5)(x-5)}{(x+1)(x-1)} \div \frac{x(x+5)-1(x+5)}{x(x-8)-1(x-8)}$$

$$= \frac{(x-9)(x+1)}{(x-9)(x-8)} \times \frac{(x+5)(x-5)}{(x+1)(x-1)} \div \frac{(x+5)(x-1)}{(x-8)(x-1)}$$

$$= \frac{x+1}{x-8} \times \frac{(x+5)(x-5)}{(x+1)(x-1)} \times \frac{(x-8)(x-1)}{(x+5)(x-1)}$$

$$= \boxed{\frac{x-5}{x-1}} \text{ Ans}$$

Question 8

$$\frac{1}{2x-3y} - \frac{x+y}{4x^2-9y^2}$$

$$= \frac{1}{2x-3y} - \frac{x+y}{(2x)^2-(3y)^2}$$

$$= \frac{1}{2x-3y} - \frac{x+y}{(2x+3y)(2x-3y)}$$

$$= \frac{(2x+3y)-(x+y)}{(2x-3y)(2x+3y)}$$

$$= \frac{2x+3y-x-y}{(2x-3y)(2x+3y)}$$

$$= \boxed{\frac{x+2y}{(2x-3y)(2x+3y)}} \text{ Ans}$$

Question 9

$$\frac{1}{x(x-y)} + \frac{1}{y(x+y)}$$

$$= \frac{y(x+y)+x(x-y)}{x(x-y)\,y(x+y)}$$

$$= \frac{xy+y^2+x^2-xy}{xy(x-y)(x+y)}$$

$$= \frac{x^2+y^2}{xy(x^2-y^2)}$$

$$= \boxed{\frac{x^2+y^2}{xy(x^2-y^2)}} \text{ Ans}$$

Question 10

$$\frac{5x+5}{3(2x-1)} + \frac{6-2x}{2(1-2x)}$$

Since $1-2x = -(2x-1)$:

$$= \frac{5x+5}{3(2x-1)} - \frac{6-2x}{2(2x-1)}$$

$$= \frac{2(5x+5)-3(6-2x)}{6(2x-1)}$$

$$= \frac{10x+10-18+6x}{6(2x-1)}$$

$$= \frac{16x-8}{6(2x-1)}$$

$$= \frac{8(2x-1)}{6(2x-1)}$$

$$= \boxed{\frac{4}{3}} \text{ Ans}$$

Question 11

$$\frac{2a}{2a-3} - \frac{5}{6a+9} - \frac{4(3a+2)}{3(4a^2-9)}$$

$$= \frac{2a}{2a-3} - \frac{5}{3(2a+3)} - \frac{4(3a+2)}{3[(2a)^2-(3)^2]}$$

$$= \frac{2a}{2a-3} - \frac{5}{3(2a+3)} - \frac{4(3a+2)}{3(2a+3)(2a-3)}$$

LCM of denominators $= 3(2a-3)(2a+3)$

$$= \frac{2a \times 3(2a+3) - 5(2a-3) - 4(3a+2)}{3(2a-3)(2a+3)}$$

$$= \frac{12a^2+18a-10a+15-12a-8}{3(2a-3)(2a+3)}$$

$$= \boxed{\frac{12a^2-4a+7}{3(2a-3)(2a+3)}} \text{ Ans}$$

Question 12

$$\frac{5}{5+x-18x^2} - \frac{2}{2+5x+2x^2}$$

Rough work: $5 \times (-18) = -90 \to 10, -9;$ $\quad 2 \times 2 = 4 \to 4, 1$

$$= \frac{5}{5+10x-9x-18x^2} - \frac{2}{2+4x+x+2x^2}$$

$$= \frac{5}{5(1+2x)-9x(1+2x)} - \frac{2}{2(1+2x)+x(1+2x)}$$

$$= \frac{5}{(1+2x)(5-9x)} - \frac{2}{(1+2x)(2+x)}$$

$$= \frac{5(2+x)-2(5-9x)}{(1+2x)(5-9x)(2+x)}$$

$$= \frac{10+5x-10+18x}{(1+2x)(5-9x)(2+x)}$$

$$= \boxed{\frac{23x}{(1+2x)(5-9x)(2+x)}} \text{ Ans}$$

Question 13

$$\frac{1-p^2}{1+q} \times \frac{1-q^2}{p+p^2} \times \left(1+\frac{p}{1-p}\right)$$

$$= \frac{(1+p)(1-p)}{1+q} \times \frac{(1+q)(1-q)}{p(1+p)} \times \left(\frac{1-p+p}{1-p}\right)$$

$$= \frac{(1+p)(1-p)}{1+q} \times \frac{(1+q)(1-q)}{p(1+p)} \times \frac{1}{1-p}$$

$$= \boxed{\frac{1-q}{p}} \text{ Ans}$$